

Myocardial Infarction



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Out line

Definition

Etiology

Pathophysiology

Types

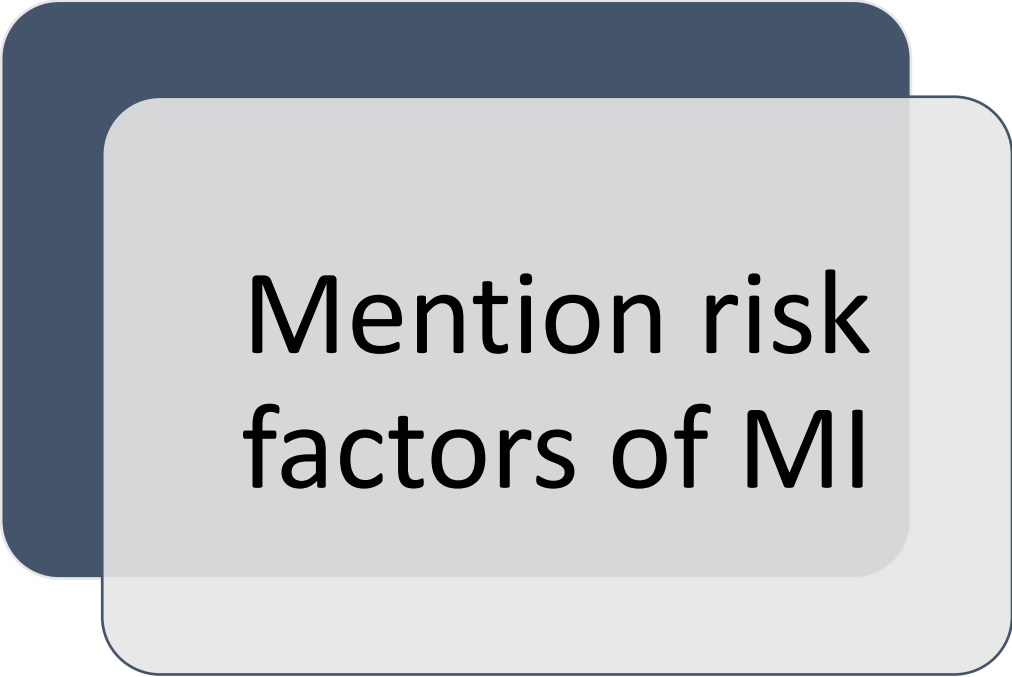
Risk factors

Signs and
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Diagnostic test

Management of
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Pre test



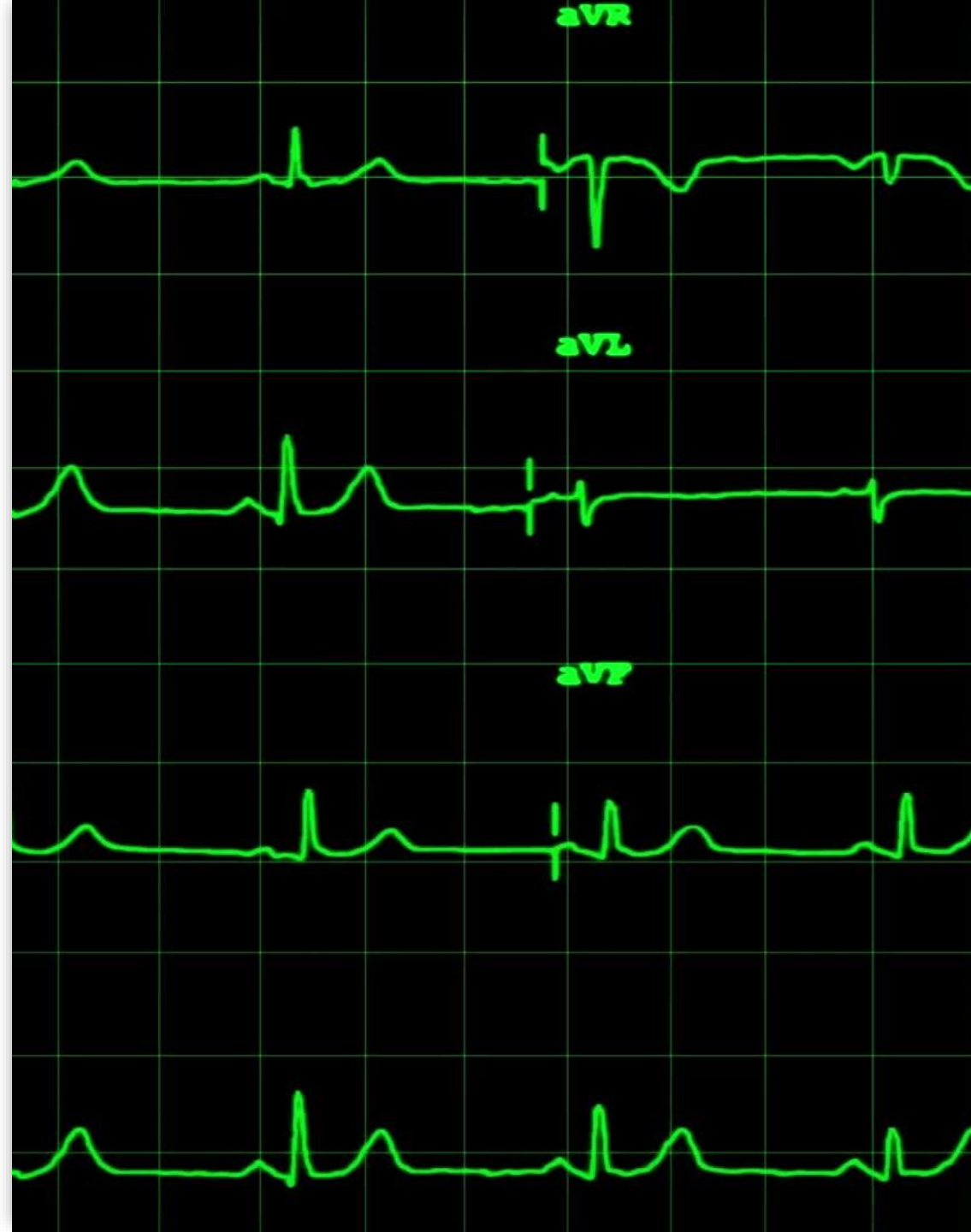
Mention risk
factors of MI



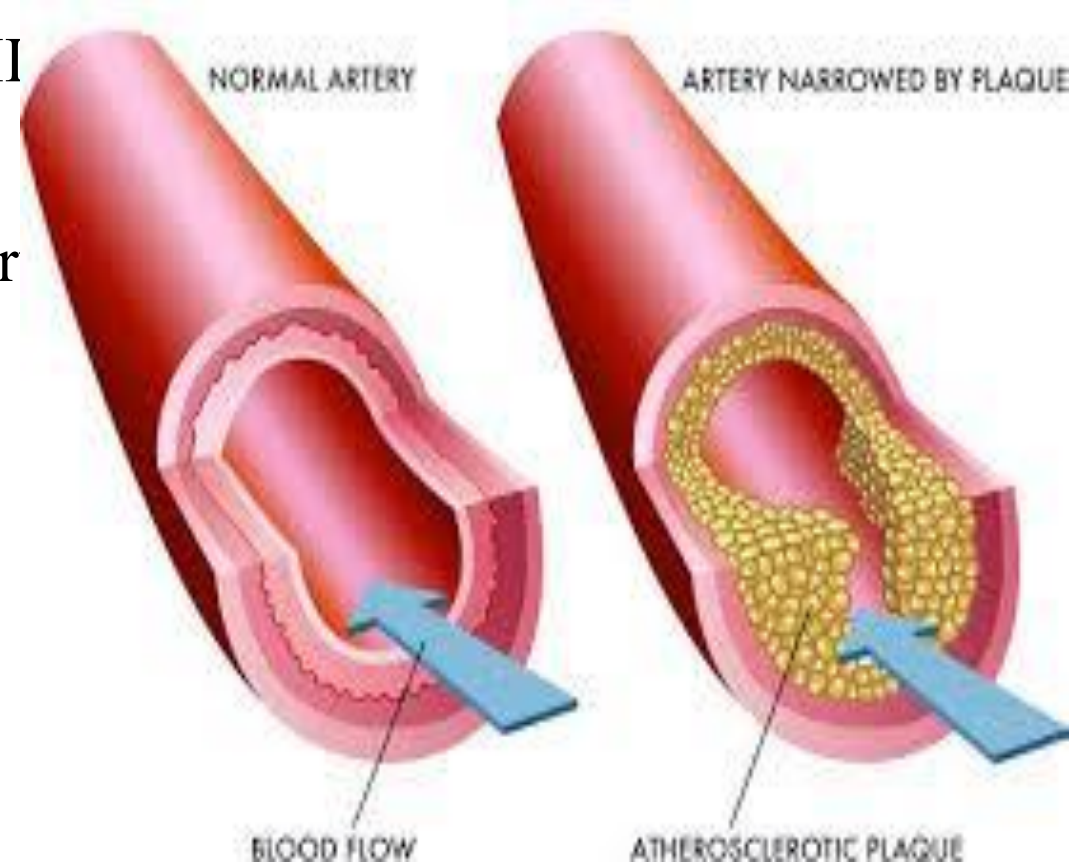
Mention clinical
manifestation
of MI

Definition:

- MI is the term used to describe an irreversible cellular loss and myocardial necrosis (cell death) secondary to an abrupt decrease or total cessation of coronary blood flow to a specific area of the myocardium.

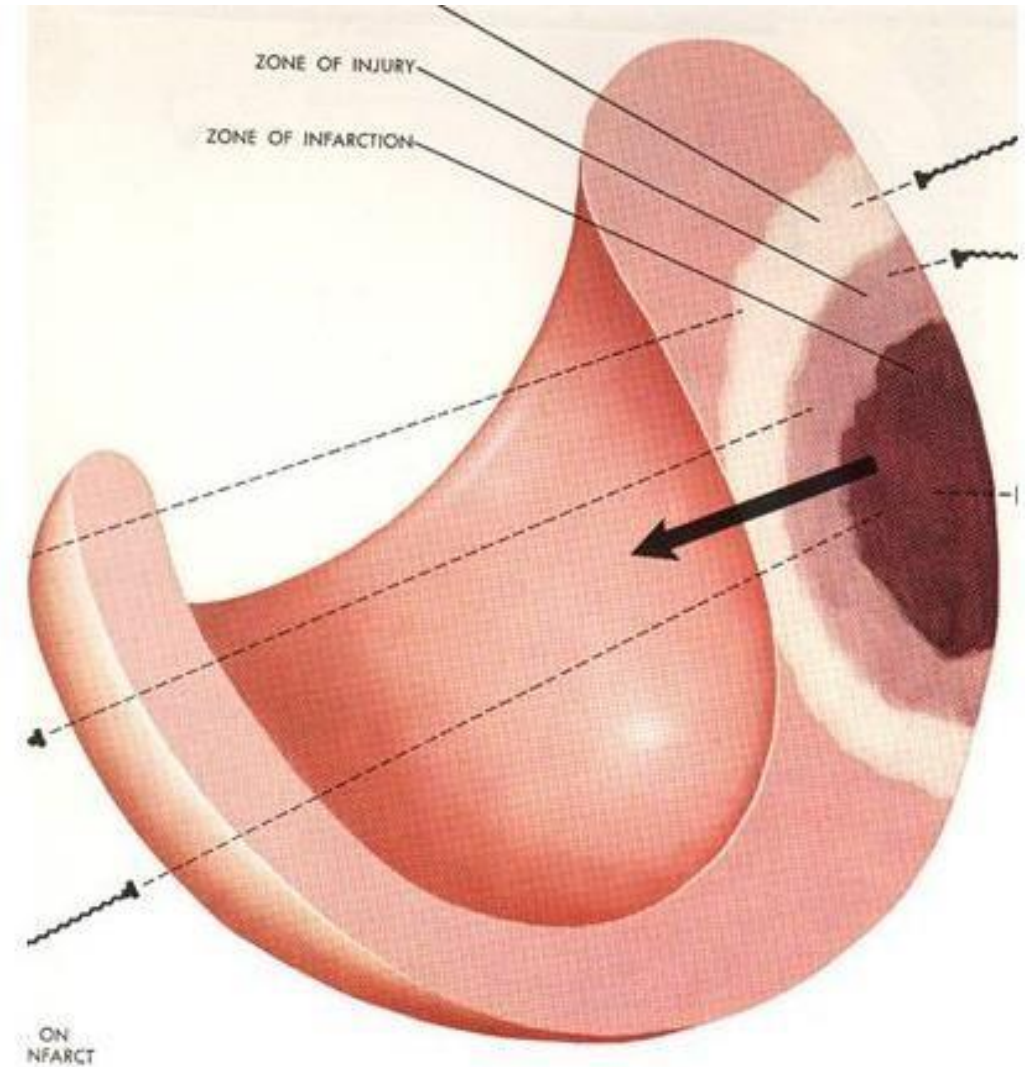


- **Etiology:**
- **Atherosclerosis** is responsible for majority of MI because it cause luminal narrowing and reduced blood flow, resulting in decreased oxygen deliver to the myocardium.
- There are three mechanisms that can block the coronary artery and responsible for the acute reduction of oxygen delivery to myocardial:
 - Plaque rupture
 - New coronary artery thrombosis
 - Coronary artery spasms to the rupture plaque



Pathophysiology:

- **Ischemia:**
 - The outer region of the infarcted myocardial area is the zone of ischemia or It is composed of viable cells.
- **Injury:**
 - The infarcted zone is surrounded by injured but still potentially viable tissues in an area known as the zone of injury.
 - This recorded on ECG as elevation of the (ST segment).

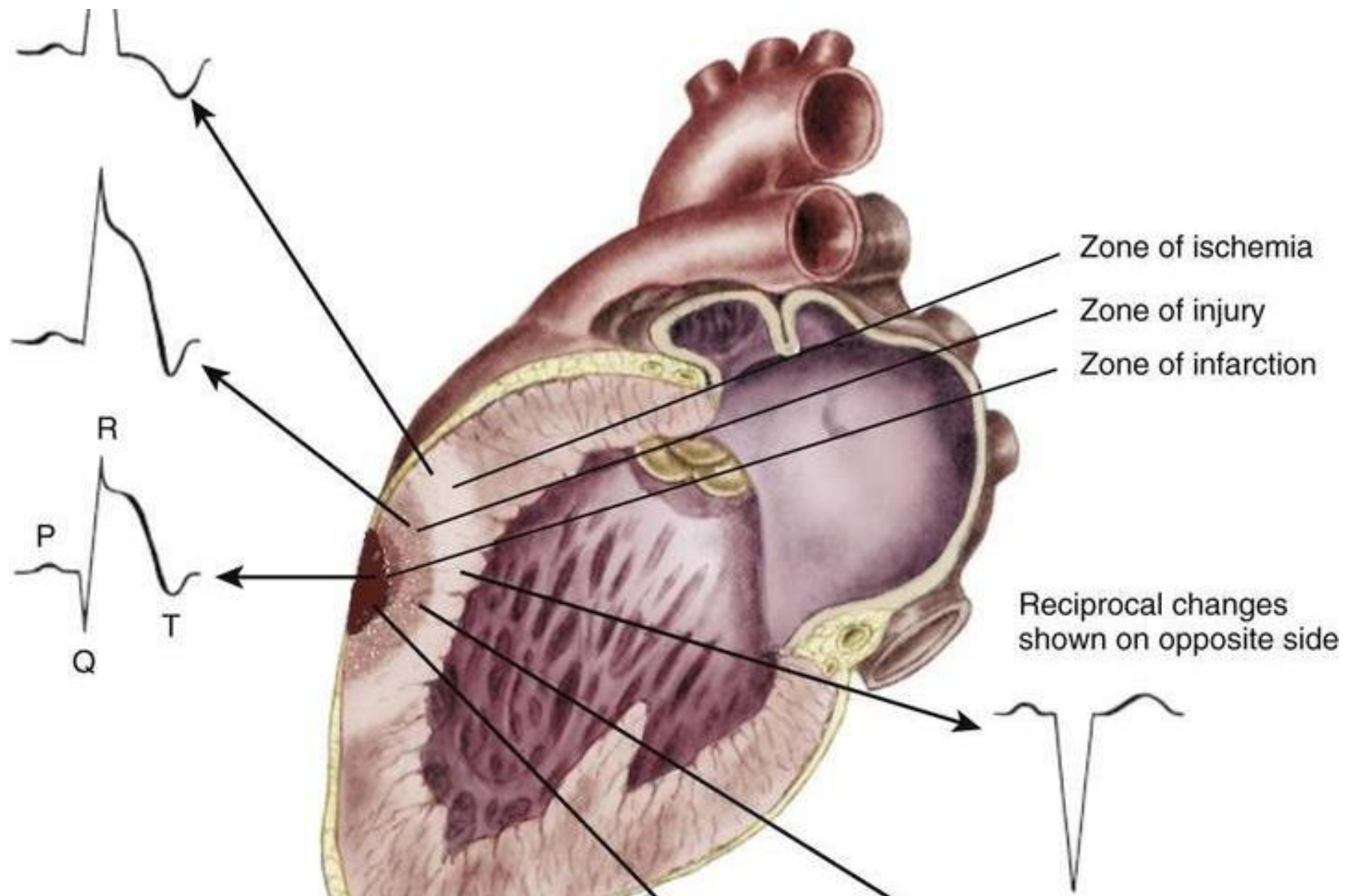


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Infarction

The area of dead and muscle (necrosis) in the myocardium is known as the zone of infarction.

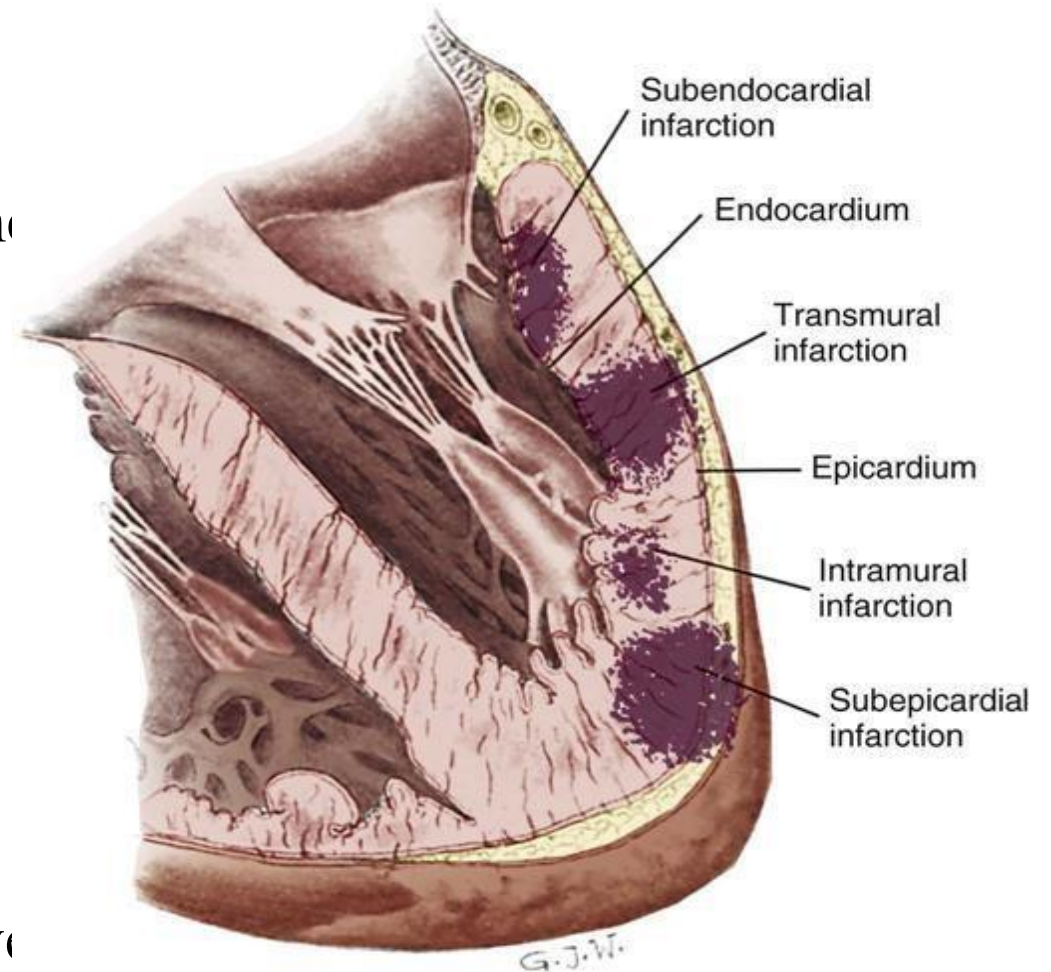
On the ECG, evidence of this zone is seen as new pathologic Q waves, which reflect a lack of depolarization from cardiac surface



Classification of infarctions:

1- A transmural infarctions:

- Involves all three muscle layers, the endocardium, the myocardium and the epicardium and result in full thickness necrosis, have a higher incidence of LV dysfunction.
- ECG demonstrates alteration in both myocardium depolarization and repolarization (ST segment). The change in depolarization is represented a new Q wave



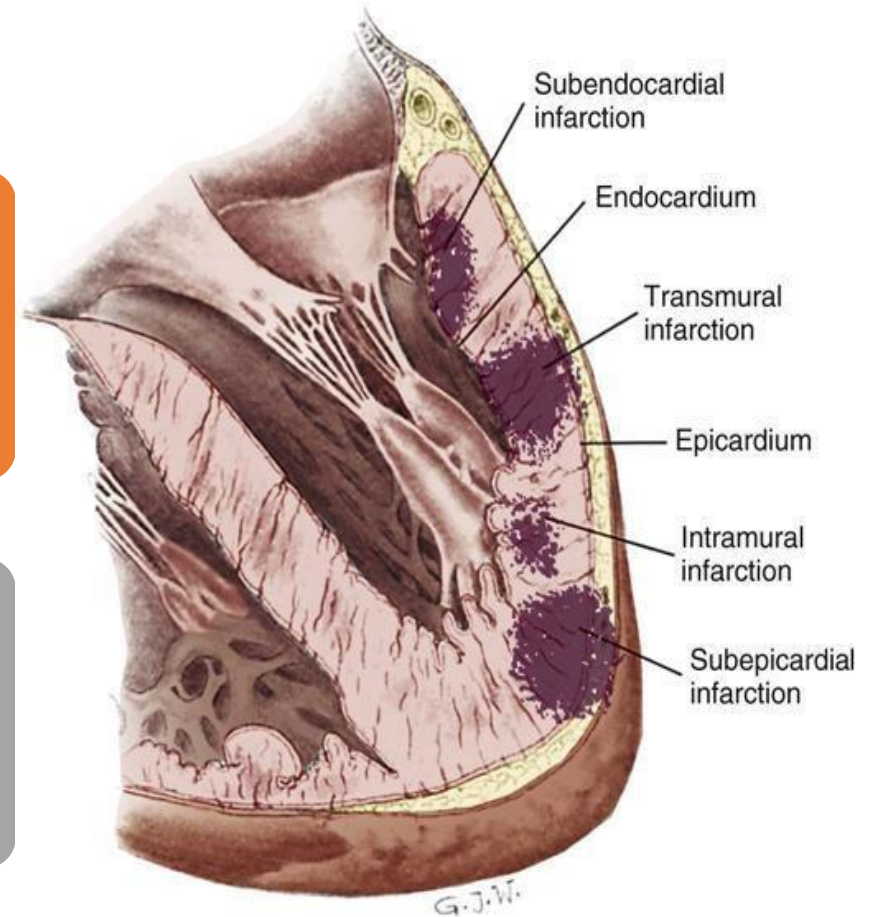
2- Non-transmural infarction:



Classified into subendocardial (involve the endocardium and myocardium) or subepicardial (involve the myocardium and epicardium).



ECG change in subendocardial ischemia is ST depression, abnormal Q wave not seen and R wave progression is normal



Myocardial Infarction Location:



Anterior Wall Infarction:



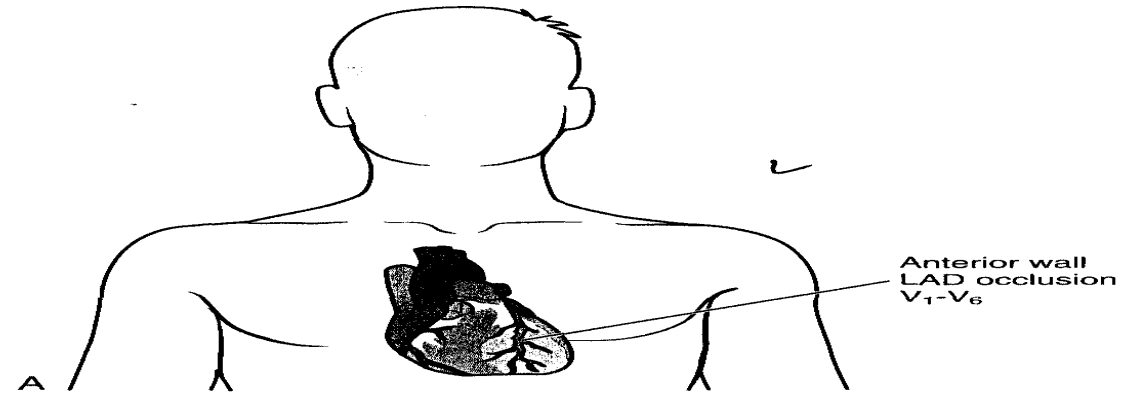
Anterior wall infarction results from occlusion of the proximal left anterior descending artery.



ST segment elevation is expected in leads V1 through V4 on the ECG.



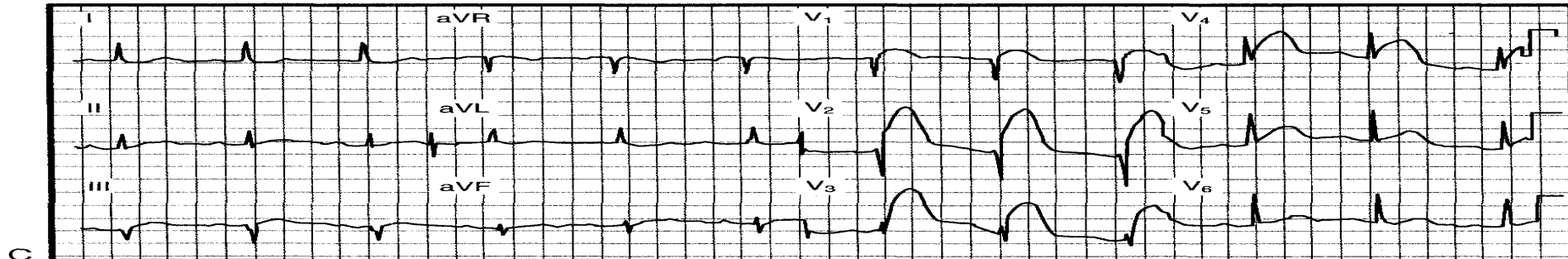
If the left main coronary artery is occluded, the ECG manifestation will involve almost all of the chest leads V1 through V6 and leads I and aVL.



LIMB LEADS **PRECORDIAL LEADS**

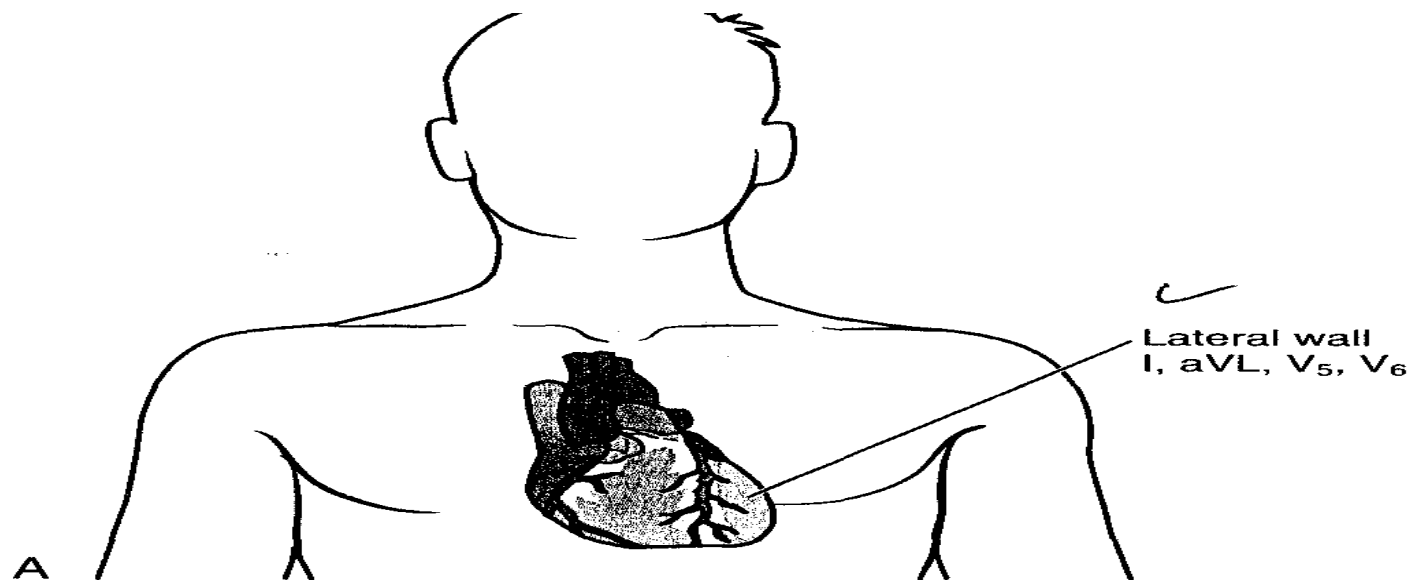
Lead I	AV _R	V ₁	V ₄
Lead II	AV _L	V ₂	V ₅
Lead III	AV _F	V ₃	V ₆

Example of an Acute Anterior Wall MI



Left Lateral Wall Infarction:

- Left lateral wall infarction occurs as a result of occlusion of the circumflex coronary artery.
- On ECG, new Q waves and ST segment and T wave changes are seen in lead I, aVL, V5 and V6.
- Patients present with only lateral wall ECG changes, and some anterior wall leads (V3 and V4) may show evidence of injury or infarction.



LIMB LEADS

PRECORDIAL LEADS

Lead I 	AV _R	V ₁	V ₄
Lead II 	AV _L 	V ₂	V ₅
Lead III 	AV _F	V ₃	V ₆

B

Inferior Wall Infarction:



Inferior wall infarction occurs with occlusion of the right coronary artery.



This infarction manifested by ECG changes in leads II, III and aVF.



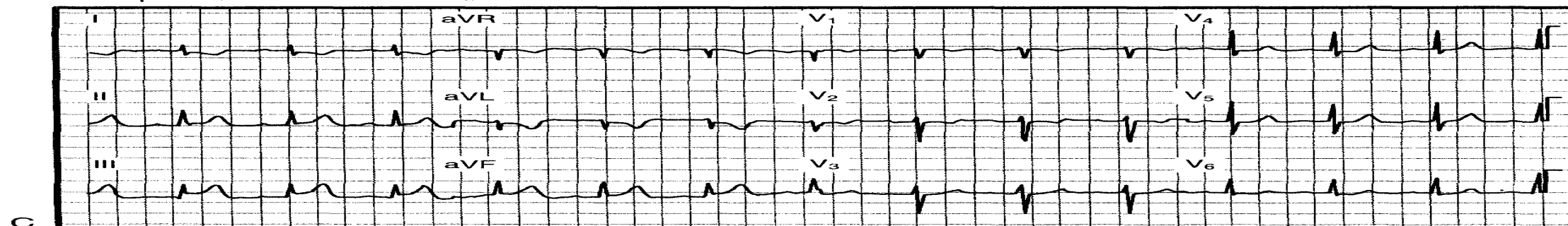
Conduction disturbances are expected with an inferior wall MI and are related to the anatomy of the coronary artery supply.

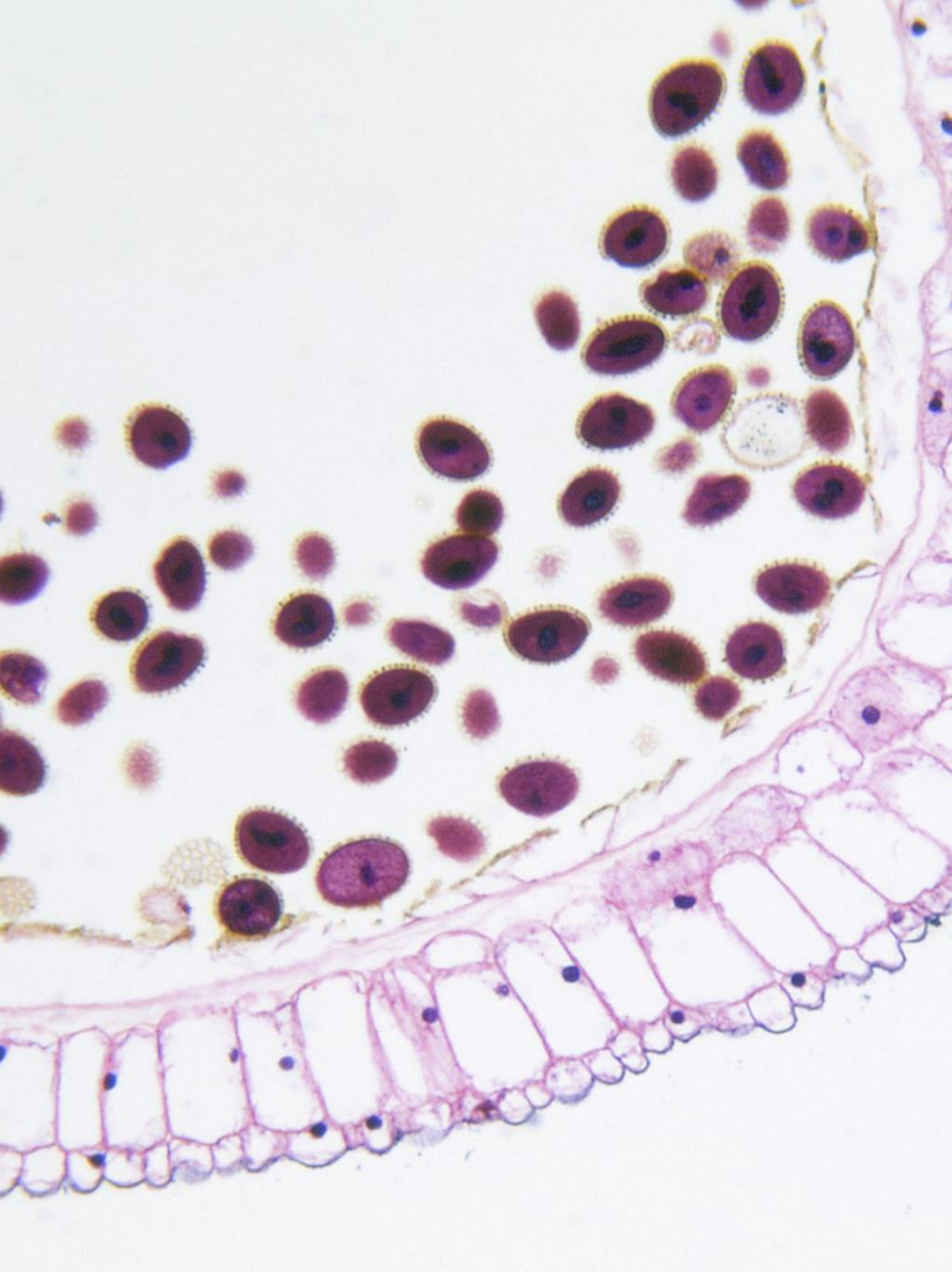


LIMB LEADS		PRECORDIAL LEADS	
Lead I	aV _R	V ₁	V ₄
Lead II	aV _L	V ₂	V ₅
Lead III	aV _F	V ₃	V ₆

B

Example of an Acute Inferior Wall MI





Right Ventricular infarction:

- Infarction in right ventricular occurs when a blockage occurs in a proximal section of the right coronary artery.
- 10% to 15% show the hemodynamic abnormalities associated with classic infarction of the right ventricle.
- It is important to write R on the ECG (V1R and V6R) in the front of the recorder right ventricular chest leads to ensure that the lead location is clear.

Posterior Wall Infarction:



Infarction of posterior wall can occur because of a blockage in the right coronary artery or in the circumflex artery.



This occurs because both arteries supply this section of the heart, although the right coronary artery is generally the dominant vessel. A posterior wall MI is difficult to detect but may be identified by specific leads V1 and V2.

Non- ST segment Elevation Myocardial Infarction:

The ECG was considered the gold standard when diagnosis acute MI.

One reason for the lack of ST segment elevation may be that the infarction and subsequent necrosis are not full –thickness lesions.

Because some of the muscle in the area can still be depolarized, ST segment elevation may not occur.

This situation is diagnosed as NSTEMI.

Risk factors

Age.

- Men age 45 and older and women age 55 and older are more risk to have a heart attack than are younger men and women.

Tobacco use.

- This includes smoking and long-term exposure to smoke that lead to atherosclerosis and plaque obstruction.

High blood pressure.

- Over time, high blood pressure can damage arteries that lead to atherosclerosis and plaque obstruction.

Risk factors

High cholesterol or triglycerides.

A high level of low-density lipoprotein (LDL) cholesterol is most likely to narrow arteries also increases heart attack risk.

Obesity.

Obesity is linked with high blood pressure, diabetes, high levels of triglycerides and bad cholesterol, and low levels of good cholesterol narrow arteries also increases heart attack risk.

Diabetes.

High blood sugar increases the risk of a heart attack.

Risk factors

Unhealthy diet.

A diet high in sugars, animal fats, processed foods, trans fats and salt increases the risk of heart attacks.

Stress.

Emotional stress, such as extreme anger, may increase the risk of a heart attack.

Clinical Manifestation of MI

Chest pain is sever, crushing, tight, squeezing or feeling of pressure

It can be precordial, substernal, or in the back, radiating to arms, neck or jaw, unrelated to exertion and respiration

The skin may be cool, clammy, pale and diaphoretic, the colour may be dusky and hyperthermia may be present

The patient may be short of breath, dyspnoea, and tachypnoea, or feel faint

Hypotension, dysrhythmia, as PVC, bradycardia or tachycardia or heart blocking with varying degree

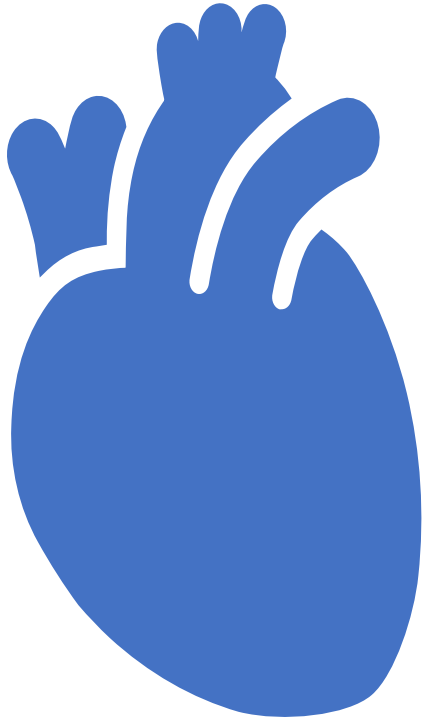
Anxious or restless or nausea and vomiting

Diagnosis of MI

- ECG Change in MI include:
 - Inverted T waves
 - ST Segment elevation
 - Pathologic Q waves



- ***Creatine kinase (CK)***
- Is an enzyme that is found in heart muscle, skeletal muscle, and brain tissue.
- CK values start rising 4 to 6 hours after the onset of myocardial necrosis, typically peak at 18 to 24 hours, and return to baseline at 36 to 40 hours.
- To rule out an MI, blood samples may be drawn every 8 hours for 24 hours.



CK-MB

- CK-MB in amounts greater than 5% of total CK generally is considered diagnostic for myocardial damage in the presence of chest pain or other symptoms believed to represent myocardial ischemia.
- Negative values 10 to 12 hours after the onset of pain indicate the absence of myocardial necrosis.
- Patients who present more than 24 hours after the onset of symptoms may not benefit from measurement of CK isoenzymes because the levels already may have returned to normal

- *Troponins:*

- Troponins are cardiac proteins that are released into the circulation after necrosis and rupture of myocardial cells.
- Three subforms have been identified: cardiac troponin-I (cTnI), troponin-T (cTnT), and troponin-C (cTnC).

Serum Marker

Biomarker	Range of Times to Initial Elevation	Mean Time to Peak Elevations (Nonreperfused)	Time to Return to Normal Range
Frequently used in clinical practice			
CK-MB	3–12 h	24 h	48–72 h
cTnl	3–12 h	24 h	5–10 d
cTnT	3–12 h	12 h to 2 d	5–14 d

Complication of Acute MI:

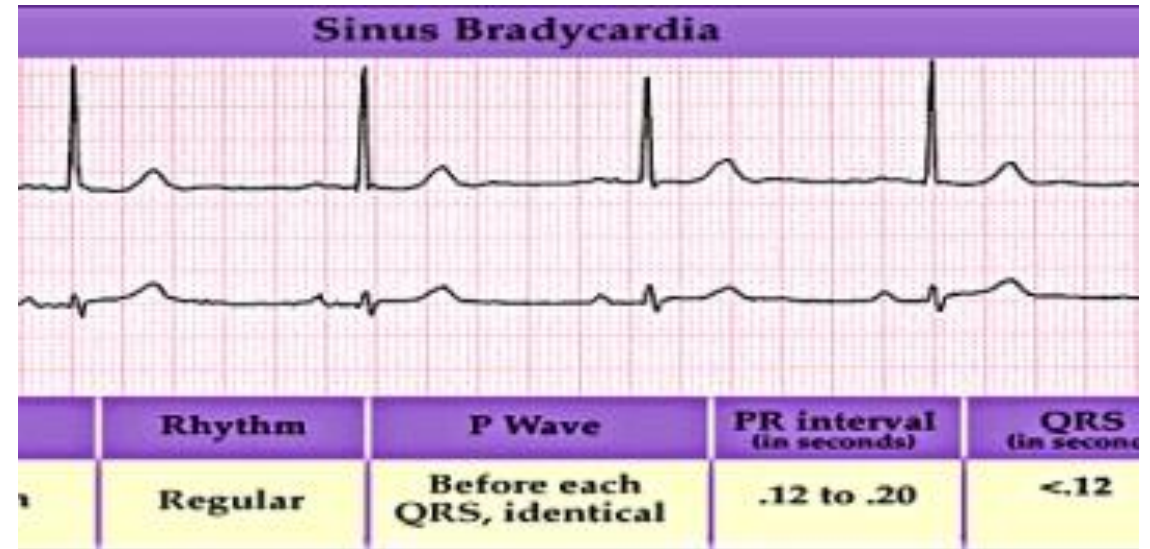


- The complication may result from electrical dysfunction or from a cardiac contractility problem.
- **Electrical dysfunctions include:**
 - Bradycardia
 - Bundle branch blocks
 - Varying degree of heart block
- **Pumping complications causes:**
 - Congestive Heart failure
 - Pulmonary edema
 - Cardiogenic shock

Electrical dysfunctions

• **Sinus Bradycardia:**

- Sinus bradycardia occur in 40% of all patients with acute MI.
- It is more prevalent with an Inferior wall MI.
- Sinus bradycardia is most frequently seen in the immediate post infraction period.
- Symptomatic bradycardia with hypotension and low cardiac output is treated with Atropine
 - 0.5 mg to 1.0 mg IV push, repeated every 3 to 5 minutes to a maximum dose of
 - 0.03 mg per kg (2 mg for a person who weight 70 kg).



Sinus Tachycardia:



Sinus tachycardia most often occurs with Anterior wall MI.

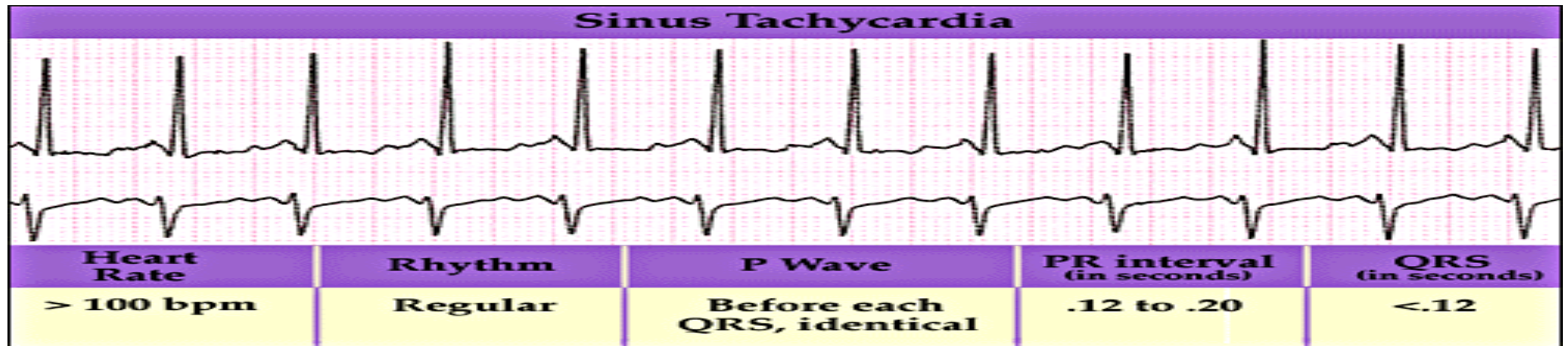
Anterior infarction impairs left ventricular pumping ability thereby reducing ejection fraction and decreasing stroke volume.



In attempt to maintain Cardiac Output, the heart rate increases. Tachycardia must be corrected because shorten diastolic filling time, thereby decreasing stroke volume, systemic perfusion, and coronary artery filling

Treatment of Sinus Tachycardia:

- If ventricular rate is greater than 150 BPM , Prepare for immediate cardioversion
- If ventricular rate is less than 150 BPM, Give Lidocaine 1.0-1.5 mg/kg IV push and repeat every 5-10 min.
- Additional doses of 0.5-0.75 mg/kg IV push are given until a maximum total dose of 3 mg/kg

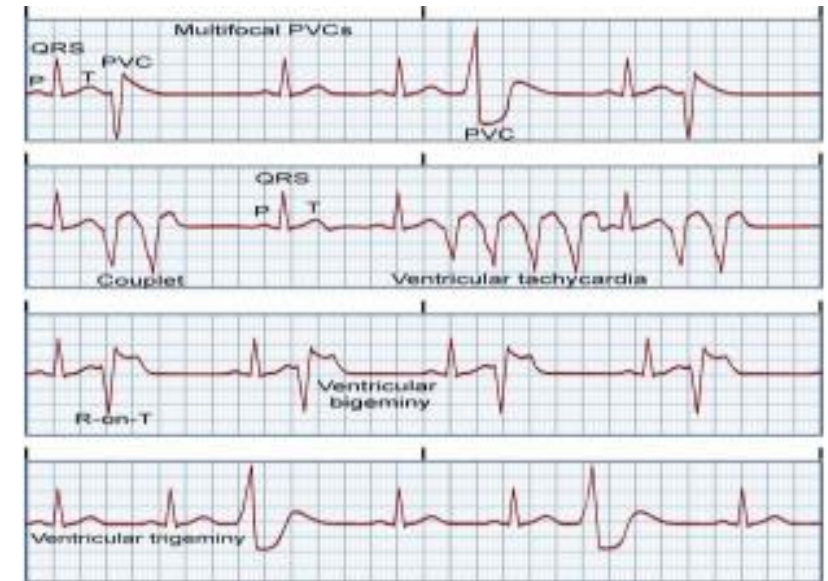
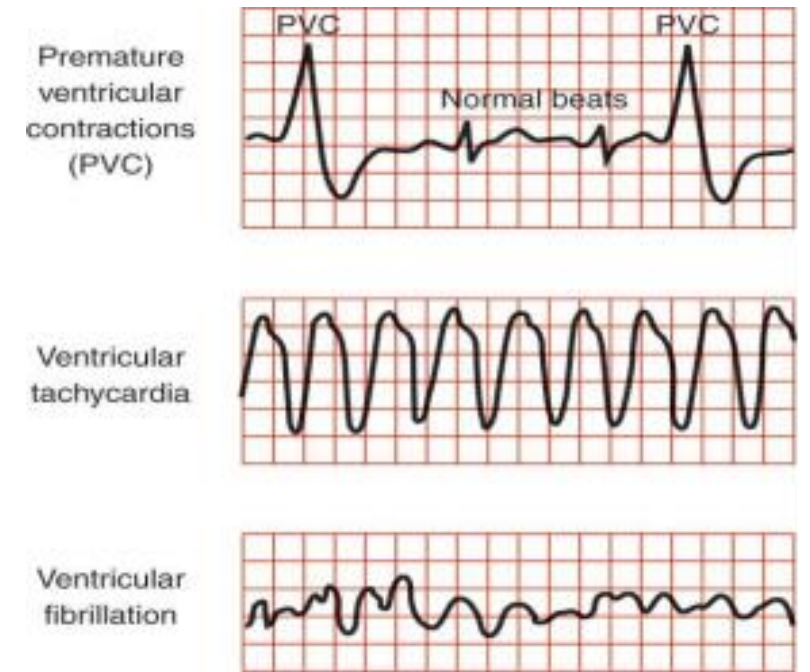


- Premature ventricular contractions**

- Within the first few hours after MI almost all patients' 'ill have Premature Ventricular Contractions (PVCs).
- PVCs and VT occurring early within the few first hours and should be treated if it is frequent (more than 6 per minute, closely coupled or multiform shape)

- PVCs are controlled by:**

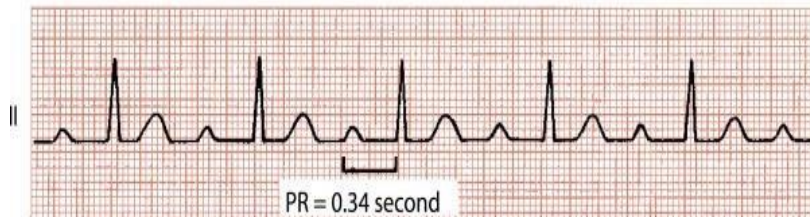
- Administering oxygen to reduce myocardial hypoxia,
- Correcting acid-base or electrolyte imbalances
- Administering an IV Lidocaine bolus and infusion



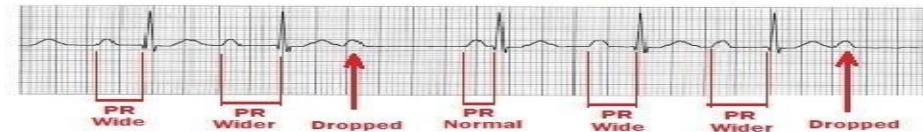
- **AV Heart Block during MI:**

- Av heart block most frequently following an inferior wall MI.
- Because the Right Coronary Artery (RCA) perfuses the AV node in 90% of the population, occlusion of RCA leads to ischemia and infarction of the AV node cells.
- Patients have AV block with hemodynamic compromise is treated by insertion of a transvenous temporary pacemaker.

First-Degree AV Block



2nd Degree AV Block Type I (Wenkebach)



1. The P-R interval successively widens until QRS fails (drops)
2. The ventricular rhythm is irregular
3. The Atrial rhythm is usually regular

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Pumping complications

Ventricular aneurysm after myocardial infarction: A left ventricular aneurysm is a non contractile, thinned left ventricular wall that results from an acute transmural infarction.

This complication occurs in less than 5% of patients after acute MI.

Ventricular septal rupture after MI: Post-infarction rupture of the septal wall is rare but potentially lethal complication of acute anterior wall MI.

This rare complication generally occurs in the first 24 hours after an acute MI.



Acute mitral regurgitation after MI:



Papillary muscle rupture can occur when the infarct involve the area around one of the papillary muscles results ineffective mitral valve closure, and blood is forced back into the low pressure left atrium during ventricular systole.



Cardiac wall rupture after MI:



The incidence of cardiac wall rupture has two peak times. The first occur within the first 24 hours and the second between the third and fifth post-infarction day when scavenger cell are removing necrotic debris, thinning the myocardial wall.



- **Pericarditis after MI:**

- Pericarditis is inflammation of the pericardial sac.
- It can occur during or after an acute MI. Pain is the most common symptoms of pericarditis and pericardial friction rub is the most common initial sign.

- **Heart failure after MI:**

- Many patients with acute STEMI have heart failure on admission.

Management of MI

- **Recanalization of the Coronary Artery**
 - Fibrinolytic Therapy
 - Anticoagulation therapy
- **Prevention of Ventricular Remolding**
 - The nurse assesses the patient's vital signs and mental status frequently.
 - The patient is placed on a cardiac monitor for ischemia and dysrhythmia detection.
 - The patient is placed on bed rest until stabilized to minimize oxygen demands.
- **Supplemental oxygen** may be given to unstable patients to increase oxygen supply.
- A pulse oximeter and arterial blood gases are used to evaluate oxygenation status.

Pain control

Pain control is a priority because continued pain is a symptom of ongoing ischemia, which increases the risk on non-infarcted myocardial tissue.

Morphine is the analgesic of choice; it decreases anxiety, restlessness, autonomic nervous system activity, and preload, thereby decreasing myocardial oxygen demands



**Administer
Nitroglycerin**



**Fibrinolytic
Therapy**



Anticoagulant therapy



**Streptokinase
is given 1.5
million U over
one hour
infusion.**



**The fibrinolytic
therapy is used
in combination
with
anticoagulants
(Heparin) and
antiplatelet
agent (aspirin).**



**IV heparin and
oral aspirin are
prescribed
during or
immediately
after
thrombolytic
therapy.**

Criteria for Fibrinolytic Therapy

- No more than 6 hours from the onset of chest pain
 - ST-Segment elevation on ECG or new-onset of left bundle branch block
 - Ischemic chest pain of 30 minutes duration
 - Chest pain unresponsive to SL Nitroglycerine
- Must check no conditions that might cause hemorrhage



Side Effects of Streptokinase

- **Bleeding** is the most common complications (close observation is needed during the 12 hours immediately after administration)
- **Allergic reactions** (SK is a bacterial protein may produce allergic reaction as anaphylaxis), thus steroids are often prescribed before SK administration to avoid this unwanted effect.
- **Hypotension** is associated with the rapid administration of SK

Contraindications of Fibrinolytic Therapy

Active
bleeding

Recent major
surgery

Stroke within 2
months

Markedly
elevated blood
pressure

Nursing Measures during Fibrinolytic Therapy

Avoid injections and vein puncture and arterial puncture

Apply additional pressure to the puncture sites to ensure hemostasis.

Placement of IV lines before administering thrombolytic therapy

Obtain a heparin lock to draw laboratory specimens

Give Antacids as a prophylactic if the patient complains of gastric discomfort

Instruct the patient to avoid vigorous tooth brushing and using straight- edge razors.

Anticoagulation:

- In acute phase after STEMI heparin is administered in combination with Fibrinolytic therapy to recanalize (open) the coronary artery.
- For the patient who receive Fibrinolytic therapy , an initial heparin bolus of 60 units / kg (maximum 5000 units) is given IV , followed a continuous heparin drip at 12 units /kg/hr (maximum 1000 units/hr) and maintain a PTT between 50 and 70 seconds (1.5 to 2 times) control for 48 hr or until revascularization.

Dysrhythmia Prevention

The antidysrhythmic with the best safety record after STEMI is amiodarone.

Beta- Blocker is another class of antidysrhythmic that are recommended for all patients after STEMI.

Beta- Blocker prevents ventricular dysrhythmia, lower blood pressure and prevents re-infarction, especially in patients with left ventricular dysfunction.

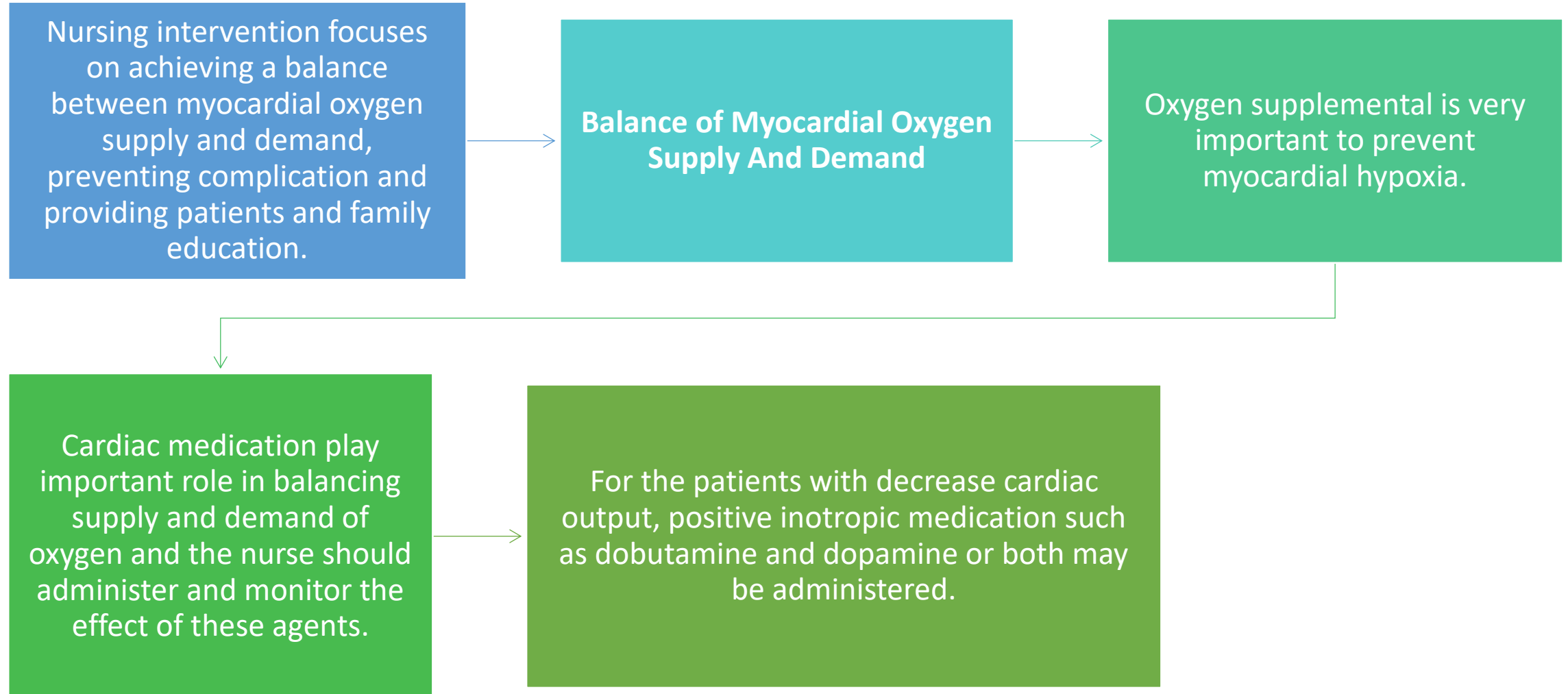
Prevention of Ventricular Remodeling:

Many patients are at risk for development of heart failure after STEMI.

Vasodilating medication (ACEIs or ARBs) can stop or limit the ventricular remodeling that lead to heart failure.

Ventricular remodeling is progressive changes in the size, architecture, and shape of the myocardium and occurs because of an injury such as an MI.

Nursing Management of MI



Prevention of complications:

The potential complications after STEMI are essential to monitor. Cardiac monitoring for early detection of ventricular dysrhythmias is essential.

It is an important nursing responsibility to monitor side effect of medication.

Hypotension is a potential complication of ACEIs, especially with the first dose. The nurse should monitor blood pressure and the patient's symptoms after taking medication.

Also, should monitor sign of bleeding if the patients receive antiplatelet, anticoagulant and fibrinolytic medication.

Nursing care

In the first 24 hours after a myocardial infarction, diet is progressed as tolerated.

Preventing hospital acquires pneumonia and DVT is facilitating by early mobilization and raising the head of the bed 30 degree or more.

An upright position facilitates decrease venous return, lowering of preload and decreasing in the workload of myocardium.

Depression after MI:

Depression is a risk factor for development of CAD and recovery following MI.

Key symptoms of depression mention frequently by cardiac patient are fatigue, change in appetite and sleep disturbance

Nursing diagnosis

- Acute Pain, related to transmission and perception of cutaneous, visceral, muscular, or ischemic impulses.
- Decreased Cardiac Output, related to alterations in preload
- Decreased Cardiac Output, related to alterations in after-load
- Decreased Cardiac Output, related to alterations in contractility
- Decreased Cardiac Output, related to alterations in heart rate or rhythm
- Activity Intolerance, related to cardiopulmonary dysfunction
- Risk for Decreased Cardiac Tissue Perfusion
- Disturbed Sleep Pattern, related to fragmented sleep
- Anxiety, related to threat to biologic, psychologic, or social integrity

Post test

- Mention classification of MI according location
- Mention complication of MI

Any question

Thank you